



Department of Computer Science and Engineering
Bachelor of Computer Science and Engineering (BCSE)
Practicum along with
Complex Engineering Problem (CEP) and Complex Engineering Activities (CEA)

- Course Code and Title: CSC 490 and Practicum
- Title of the Practicum: Online House Rent Finder System
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- How the Knowledge Profile K's (K1, K2, ...) are addressed ?

Knowledge Profile K's	Addressed (Explain how it was addressed?) or —
K1	Not Applicable. This practicum is focused on software engineering, web application development, database management, and online rental management systems, so direct application of natural sciences (physics, chemistry, biology) is not required.
K2	Addressed. Mathematical and computational concepts are applied through function point analysis, database normalization, project estimation, system complexity analysis, and structured data modeling. These techniques are used to design and analyze the Online House Rent Finder System efficiently. (Ch. 6 Project Planning; Ch. 7 Database Design)
K3	Addressed. Core engineering fundamentals are applied through the development of a full-stack system using MVC architecture, backend logic, authentication systems, role-based access control, booking management, and database handling. (Ch. 4 Analysis; Ch. 3 Requirement Engineering)
K4	Addressed. Specialized software engineering knowledge is demonstrated through full-stack web development using HTML, CSS, Bootstrap, JavaScript, PHP Laravel, MySQL, and SSLCommerz payment integration along with role-based dashboards for Admin, Owner, and



	Tenant users. (Abstract; Ch. 1 Introduction; Ch. 7 Designing)
K5	Addressed. Engineering design principles are followed through the complete software development lifecycle including requirement analysis, ERD, Use Case Diagram, Activity Diagram, implementation, testing, and deployment planning. (Ch. 3 Requirement Analysis; Ch. 7 Designing; Ch. 8 Testing)
K6	Addressed. Modern tools and technologies are used including PHP Laravel, MySQL, HTML, CSS, Bootstrap, JavaScript, XAMPP, GitHub, and SSLCommerz API integration for secure online payment handling, demonstrating practical engineering knowledge. (Ch. 3 System Requirements; Abstract)
K7	Addressed. Ethical, societal, and professional issues are considered including secure payment handling, user privacy, data protection, fairness, accessibility, and system reliability. The system also reduces paperwork and supports digital rental management services. (Ch. 9 Ethical Consideration)
K8	Addressed. Research and investigation are conducted through feasibility study, requirement analysis, risk management (RMMM), testing strategies, software documentation, and studying reference materials and online resources to improve system usability, scalability, and security. (Ch. 5 Project Management; Ch. 6 Planning; References)

- Which P's are addressed (Requirements: P1 and at least two from P2 to P7)?

Name of CEA	Addressed (Explain How?) or Not Addressed
P1: Depth of Knowledge	Addressed. The project requires strong knowledge of software engineering, full-stack web development, database management, authentication systems, online booking management, and secure payment integration. It combines theoretical knowledge with real-world implementation in a complete rental management system.
P2: Conflicting Requirements	Addressed. The system balances multiple conflicting requirements such as security vs usability (secure login and payment while maintaining a user-friendly interface), scalability vs performance, and responsiveness vs functionality for Admin, Owner, and Tenant users.



P3: Depth of Analysis	Addressed. The project involves detailed requirement analysis, ERD, Use Case Diagram, MVC architecture, risk analysis, booking validation, payment verification, and testing strategies, requiring analytical thinking and advanced problem-solving skills.
P4: Familiarity of Issues	Addressed. The system deals with real-world challenges such as secure payment gateway integration, multi-role user management, booking approval systems, notification handling, duplicate booking prevention, and secure authentication, which are beyond typical classroom-level problems.
P5: Extent of Applicable Codes	Addressed. The project follows standard software engineering practices including secure authentication, data validation, structured coding standards, role-based authorization, and ethical handling of user and payment data aligned with professional standards.
P6: Extent of Stakeholders	Addressed. Multiple stakeholders are involved including tenants, property owners, administrators, supervisors, and the organization. Their requirements are reflected in the system functionalities, workflows, and dashboard management features.
P7: Interdependence	Addressed. The system consists of interconnected modules such as frontend interfaces, backend logic, booking management, payment gateway integration, notifications, authentication systems, and databases. Changes in one component affect other modules, requiring careful system integration and coordination.



- Which A's are addressed (Requirements: At least two from A1 to A5)?

CEA [Only Relates to PO(j)]	
Name of CEA	Addressed (Explain How?) or Not Addressed
A1: Range of Resources	Addressed. The project uses various resources including development tools (PHP Laravel, MySQL, Bootstrap, JavaScript), platforms (XAMPP, GitHub), online documentation, payment gateway APIs, testing devices, and guidance from supervisors and industry professionals.
A2: Level of Interaction	Addressed. The system involves interaction between multiple components such as frontend, backend, database, payment gateway, notification system, and role-based dashboards, as well as collaboration with supervisors and real-world organizational workflows.
A3: Innovation	Addressed. The project integrates online property searching, booking requests, secure payment processing, role-based dashboards, notifications, and rental management into one unified digital platform, improving traditional manual house rental systems.
A4: Consequences to Society and Environment	Addressed. The system contributes to society by improving access to digital rental services, reducing manual paperwork, increasing transparency, and supporting secure online transactions. Ethical considerations such as user privacy and fairness are also maintained.
A5: Familiarity	Addressed. Although based on familiar technologies, the complete integration of full-stack development, online booking systems, payment gateway integration, role-based management, and real-world deployment makes it a complex and practical engineering activity beyond typical academic projects.



Comments: The practicum work successfully demonstrates the application of software engineering principles, full-stack web development, database management, authentication systems, and secure online payment integration in solving a real-world house rental management problem. The project effectively addresses Complex Engineering Problems (CEP) and Complex Engineering Activities (CEA) through requirement analysis, system design, implementation, testing, and deployment planning. The developed system is practical, scalable, user-friendly, and capable of supporting digital transformation in rental property management services.